

Rule-Based AI Chatbot Project Report

Internship Project 1 – Artificial Intelligence

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Project Objective

The objective of this project is to design and implement a simple rule-based AI chatbot using Python. The chatbot interacts with users through predefined responses and demonstrates the fundamentals of Artificial Intelligence and decision-making logic.

Tools & Technologies Used

- Python Programming Language
- Visual Studio Code (VS Code)
- Command Prompt / Terminal

Project Introduction

A rule-based chatbot is one of the simplest forms of Artificial Intelligence. It follows predefined rules and responds according to conditions written by the programmer. This project helps beginners understand AI concepts, loops, conditional statements, and user interaction.

Key Features

- Handles greetings such as Hello, Hi, and Hey
- Answers common questions
- Responds to thank-you messages
- Supports exit commands (Bye, Exit, Quit)
- Runs continuously until the user exits
- Easy to understand and beginner-friendly

Working Procedure

1. Program starts and displays a welcome message.
2. User enters a message.
3. Input is converted to lowercase.
4. Conditions are checked using if-elif-else statements.
5. Appropriate response is displayed.
6. Program continues until the user types bye, exit, or quit.

Python Source Code

```
print("Welcome to AI Chatbot!") while True: user = input("You: ").lower() if user in ["hello", "hi", "hey"]: print("Bot: Hello! Nice to meet you.") elif user == "how are you": print("Bot: I am doing great!") elif user == "what is your name": print("Bot: My name is RuleBot.") elif user == "who made you": print("Bot: I was created by an AI Intern.") elif user == "what can you do": print("Bot: I can answer simple questions.") elif user == "thank you": print("Bot: You're welcome!") elif user in ["bye", "exit", "quit"]: print("Bot: Goodbye!") break else: print("Bot: Sorry, I don't understand that.")
```

Sample Output

You: hello

Bot: Hello! Nice to meet you.

You: what is your name
Bot: My name is RuleBot.

You: bye
Bot: Goodbye!

Advantages

- Easy to develop and understand
- Fast response time
- Suitable for beginners learning AI
- Requires minimal resources

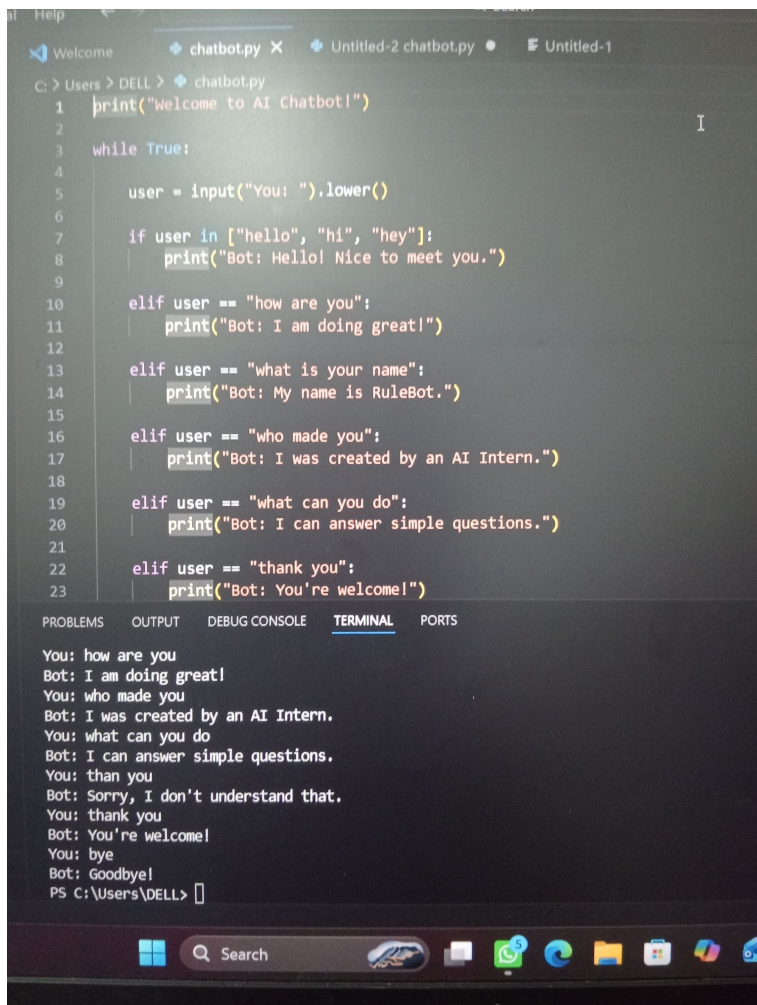
Future Improvements

The chatbot can be enhanced using Natural Language Processing (NLP), voice recognition, machine learning, and database integration to support more intelligent conversations.

Conclusion

This project successfully demonstrates the basic concepts of Artificial Intelligence through rule-based decision making. The chatbot uses Python control flow and conditional logic to simulate human interaction and provides a strong foundation for advanced AI projects.

Output



The screenshot shows a Python chatbot program running in a code editor (VS Code) and a terminal window. The code in the editor is as follows:

```
1 print("Welcome to AI Chatbot!")
2
3 while True:
4     user = input("You: ").lower()
5
6     if user in ["hello", "hi", "hey"]:
7         print("Bot: Hello! Nice to meet you.")
8
9     elif user == "how are you":
10        print("Bot: I am doing great!")
11
12    elif user == "what is your name":
13        print("Bot: My name is RuleBot.")
14
15    elif user == "who made you":
16        print("Bot: I was created by an AI Intern.")
17
18    elif user == "what can you do":
19        print("Bot: I can answer simple questions.")
20
21    elif user == "thank you":
22        print("Bot: You're welcome!")
23
```

The terminal window shows the following output:

```
You: how are you
Bot: I am doing great!
You: who made you
Bot: I was created by an AI Intern.
You: what can you do
Bot: I can answer simple questions.
You: than you
Bot: Sorry, I don't understand that.
You: thank you
Bot: You're welcome!
You: bye
Bot: Goodbye!
PS C:\Users\DELL>
```